



phies greatly favour the cloud scheme of things. A cloud is, after all, a fluffy and lovable structure. Would we not hate a boxed cloud?

Undoubtedly, Open Source lies at the foundation of many of the earliest cloud implementations. "Open Source has built the cloud. When we think of the services we consume on the cloud, from Facebook and Google to Amazon, none would have been affordable or scalable using a traditional licensing model. Moving that capability from the leading edge of SaaS provision and into the mainstream enterprise is the next big opportunity for Open Source," says Prakash Advani, partner manager—Central Asia, Ubuntu.

The fact that most of the public clouds, including Amazon's EC2, run on Linux-based platforms, is just one facet of the story. In fact, that is now taken so much for granted that the focus has shifted to tools and platforms that enable the building of private or hybrid clouds, the integration of legacy infrastructure with the cloud, and so on.

A rather large number of such Open Source tools are now available, right from platforms and development tools, to management dashboards and automated migration tools for applications. Existing Open Source platforms are also fast adapting to the needs of cloud computing and include features such as intelligent workload management and cloud-enabled scalability considerations, to help massive hori-

zontal scalability at all the layers of the technology stack. Ubuntu, Red Hat, and almost every other Open Source platform has a stable cloud offering now.

George Paul, executive vice president, HCL Infosystems, justifies our point with some examples: "An Open Source software-infrastructure project called Eucalyptus imitates the experience of using Amazon's EC2,

but allows users to run programs on their own resources. The University of Chicago's Nimbus is another Open Source cloud-computing project that is widely recognised as having pioneered the field. Today, customers have a large choice of Open Source applications for the cloud, including Red Hat, Traffic Server, Puppet, Zoho, Cloudera, Enomaly and Joyent."

He goes on to explain that apart from new tools, existing Open Source offerings are being made cloud-enabled through standardisation. "A common standard called the application packaging standard (APS), an Open standard with all specifications, has been introduced; it helps in making applications multi-tenant. It consists of over 250 applications, and will continue to grow in the future as well," he says. APS helps to standardise the packaging, automate the provisioning and management, and to integrate with other hosted services.

HCL's O'zone, a cloud-enabled services suite, combines a variety of Open Source solutions, including the Proxmox Open Source virtualisation platform, Red Hat Enterprise Virtualisation, and Open Source enterprise resource planning, customer relationship management and content management systems.

Open Source for the cloud

Virtualisation of infrastructure is at the heart of cloud computing. Open Source options, such as the Kernel-based Virtual

Machine (KVM) for Linux and the Xen hypervisor, are very competent. Other Open Source Linux-native tools like Hadoop, Cassandra, HipHop, CouchDB and Btrfs also assist in building a first-class data centre very cost-effectively. These could be seen as the starting point for the coming wave of enterprise-scale Open Source adoption on the cloud.

Software platforms such as Eucalyptus, which enable the implementation of private and hybrid clouds, are also becoming very popular. Eucalyptus is a modular platform that is capable of working with a variety of interfaces, including Amazon's EC2 and Simple Storage Service (S3) Web services. Eucalyptus works with various distros, including Red Hat Enterprise Linux (RHEL), CentOS, SUSE Linux Enterprise Server (SLES), OpenSUSE, Debian and Fedora. It can also host MS Windows images. It is capable of working with many virtualisation technologies such as VMware, Xen and KVM hypervisors, in order to implement the abstraction demanded by a cloud environment.

Solutions such as Cloudera are built on the capabilities of popular Open Source options like Hadoop to meet an enterprise's cloud-computing needs. The Cloudera data management platform incorporates the Hadoop Distributed File System (HDFS), Hadoop MapReduce, Hive, Pig, HBase, Sqoop, Flume, Oozie, Zookeeper and Hue, and is available free under an Apache licence. The enterprise package includes support, tools and training.

Joyent's SmartPlatform, basically PaaS, is an Open Source, server-side JavaScript-based framework for developing and delivering real-time, asynchronous Web applications to the cloud. While hosting is free at the moment, it might become a paid service once SmartPlatform graduates from the beta to a stable release. People are betting big on Joyent's offering, because of the ubiquity of Javascript.

Enomaly's Open Source cloud-management and provisioning software, the Elastic Computing Platform (ECP), allows an enterprise to create a